

Networks and Security

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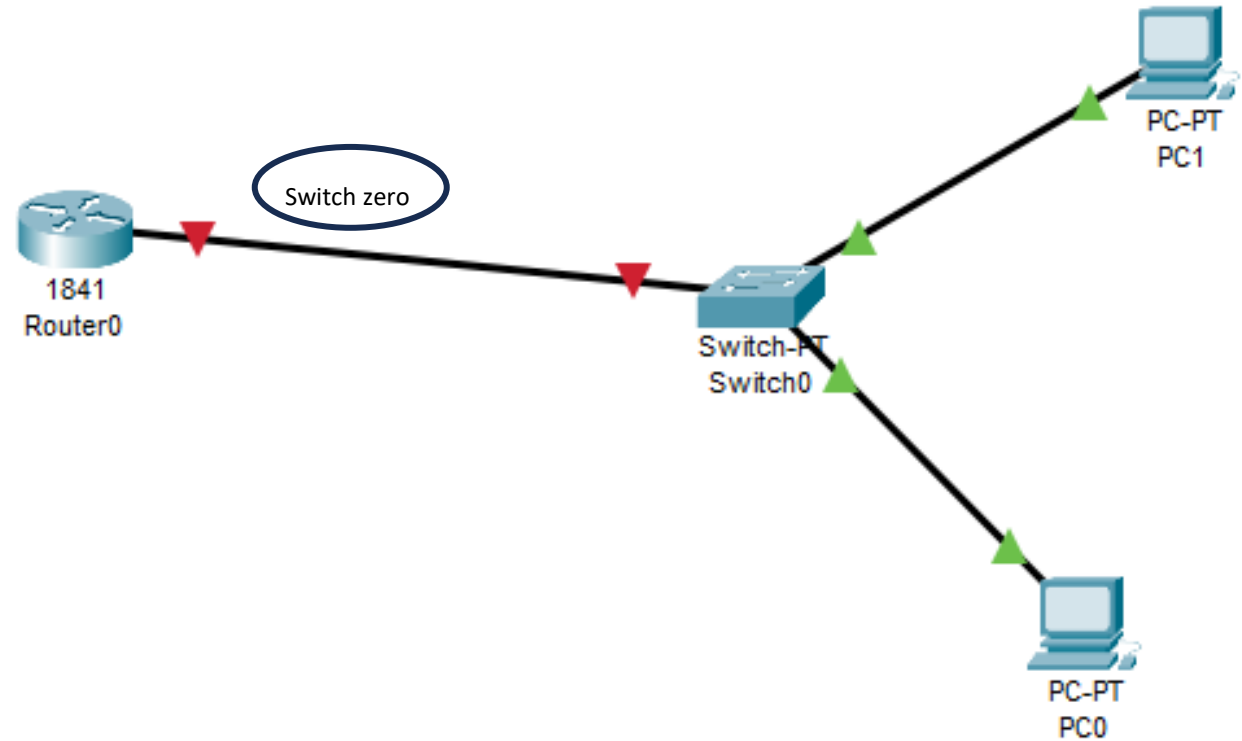
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DHCP Configuration

Set up your network

- Router (1841)
- Switch (PT-Switch)
- 2 PCs



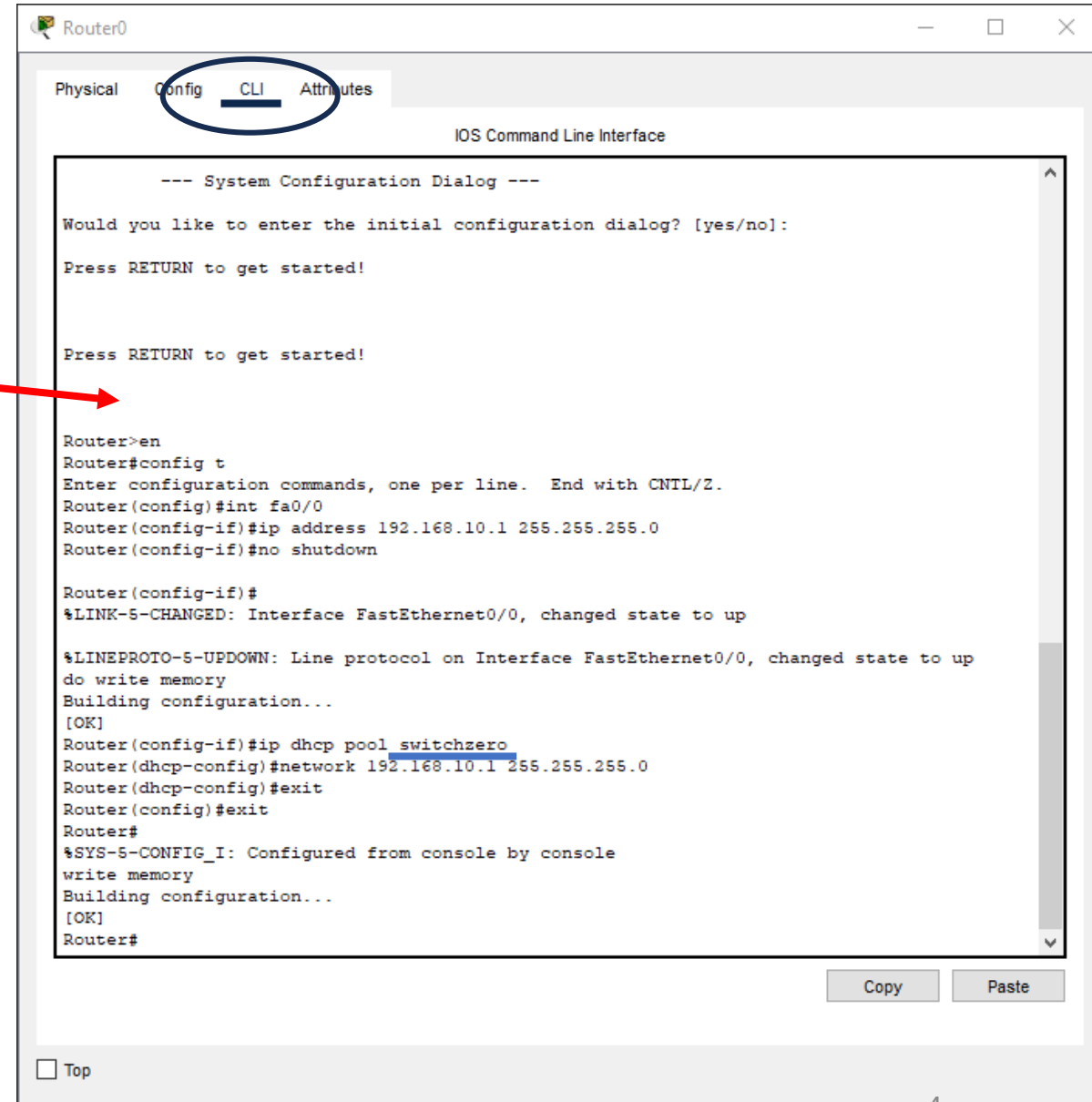
Router

On your router and from CLI,
follow the instruction provided
through the CLI window:

Router>en

Router#config t

Continue.....



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

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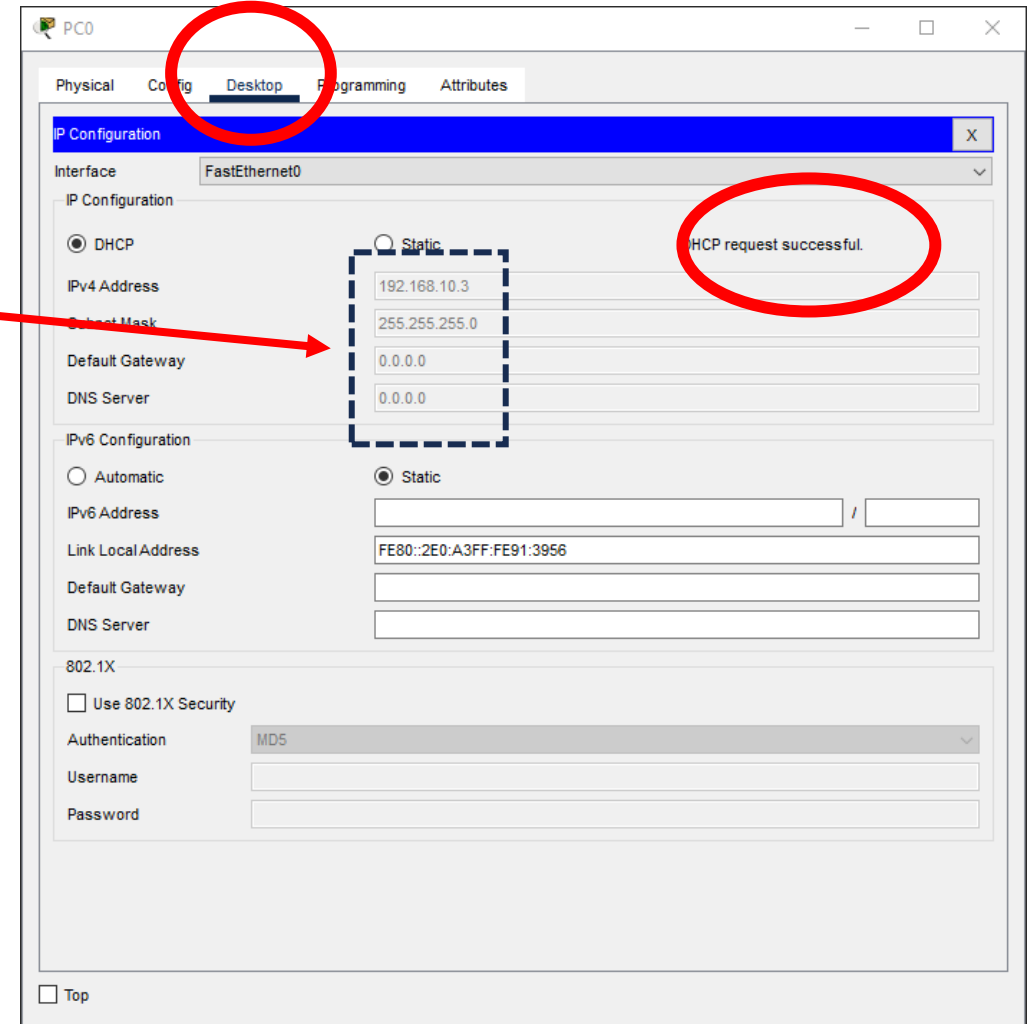
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
do write memory
Building configuration...
[OK]
Router(config-if)#ip dhcp pool switchzero
Router(dhcp-config)#network 192.168.10.1 255.255.255.0
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

PC0

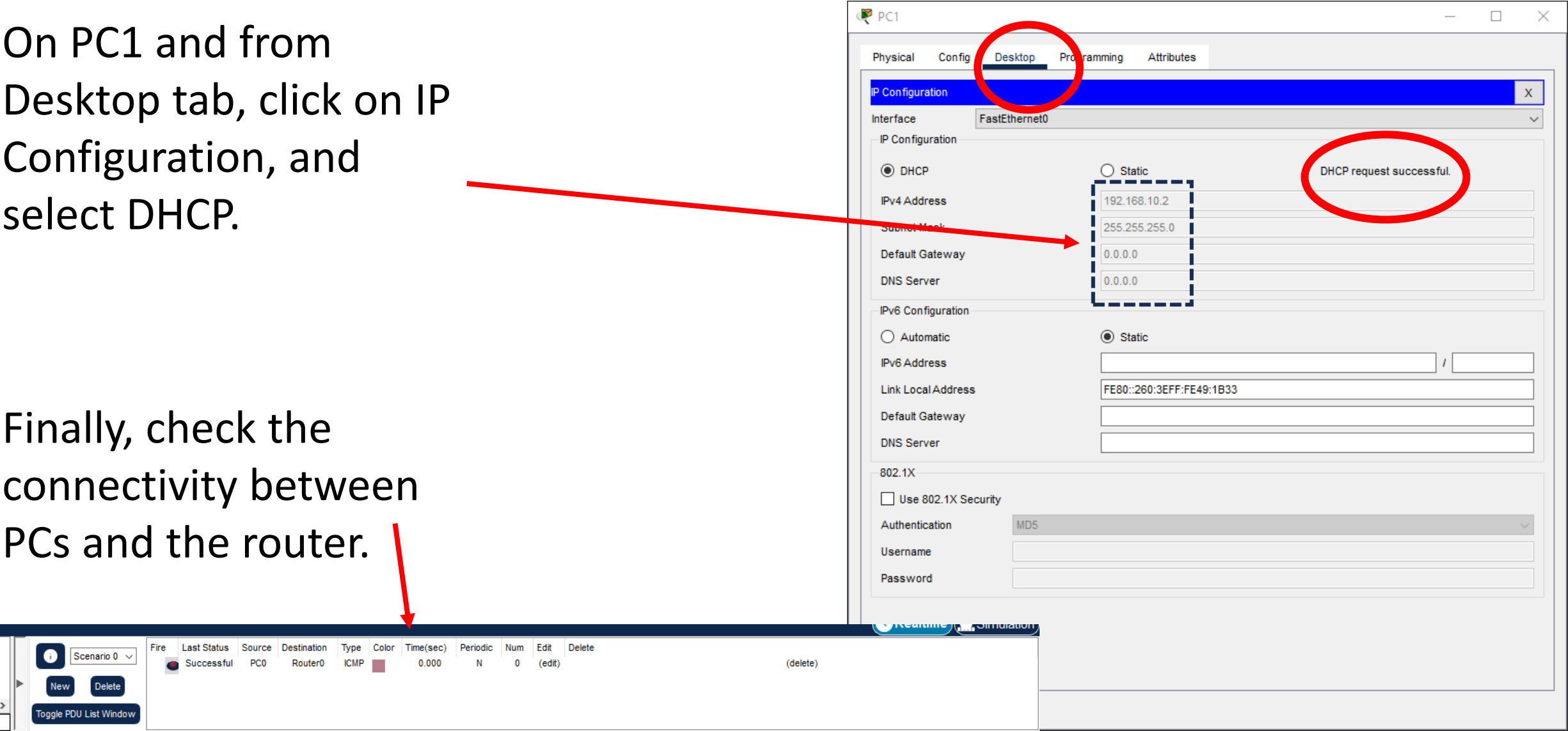
On PC0 and from Desktop tab, click on IP Configuration, and select DHCP.



PC1

On PC1 and from Desktop tab, click on IP Configuration, and select DHCP.

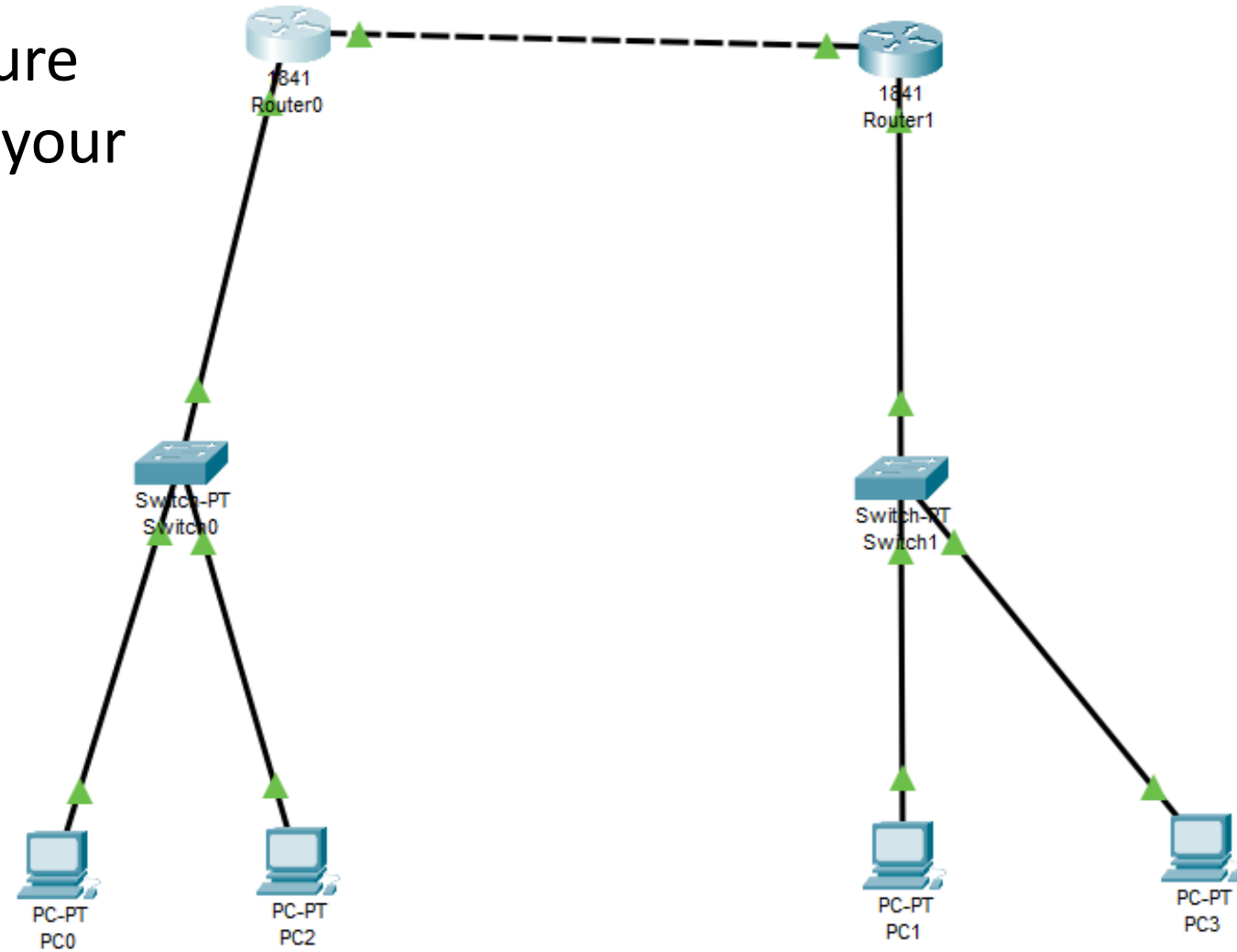
Finally, check the connectivity between PCs and the router.



Exercise

Objectives

Design the given topology and configure DHCP to demonstrate that the PCs in your network are successfully connected.



Hints

You may need to use some useful commands to set up and verify the network, such as those for configuring interfaces, and enabling DHCP. Below are some commands that might help, but you may need to do some additional research:

- *For instance, on Router 0 (R0):*
- *Exclude the router's IP address from the DHCP pool:*
- *R0(config)# ip dhcp excluded-address 192.168.10.1*
- *R0(config)# ip dhcp pool LAN1*
- *R0(dhcp-config)# network 192.168.10.0 255.255.255.0*
- *R0(dhcp-config)# default-router 192.168.10.1*
- *On Router 0 (R0):*
- *Add a static route to reach PC2's network via Router 1:*
- *R0(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2*